

ROLE OF EDUCATION AND POLITICAL VIEWS ON BELIEF UPDATING UNDER HIGH SOCIAL PREVALENCE

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Introduction

Our beliefs shape how we interact with the world. Some beliefs directly influence our decision-making, for example, evaluating claims such as “vaccines cause autism”. We often weigh the potential consequences of such statements before updating our beliefs. Other beliefs such as “the Earth is flat”, have little immediate relevance unless we work in fields that require accurate knowledge of planetary geometry for navigation or scientific decision-making. Even when we attempt to gather information, it is impossible to independently verify every claim. Some statements, like “cats have whiskers”, are easily validated through direct observation, whereas others, such as “humans have landed on the moon”, require us to rely on testimony from journalists and astronauts. For claims about events that predate modern documentation, such as “humans and dinosaurs co-existed”, we depend on expert consensus. When expert opinion is divided, as in the case of claims like “most physical ailments can be remedied through realignment of the skeletal system”, individuals face greater uncertainty.

Under such ambiguous conditions, people often rely on social prevalence when deciding whether to integrate a statement into their belief system. Because individuals aim to minimize cognitive effort, they are unlikely to search for evidence unless the claim holds personal significance. When uncertainty is high, the perceived popularity or endorsement of a claim increases the likelihood that it will be incorporated into one’s beliefs (Orticio, 2021). The present study examines how education and political orientation influence belief updating. Under high social prevalence and uncertain conditions, I predict that participants with less education show greater belief changes, whereas participants with more liberal political views show less belief change.

Methods

Participants

Four hundred and three Americans fluent in English were recruited through Prolific (prolific.co) to participate in an 18-minute online experiment. Participants were compensated at a rate of \$10/hour, with an opportunity to receive a bonus for good performance. This data was collected for the paper “Beliefs are most swayed by social prevalence under uncertainty” by Evan Orticio, Louis Martí, and Celeste Kidd.

The dataset used for the final project includes a subset of 205 participants (163 female, 34 male, 7 non-binary; age range: 18-72; 144 White, 21 Black or African American, 16 Multiracial, 14 Asian, 7 Other, 2 American Indian or Alaska Native).

Measures/Materials

The 37 real-world conspiratorial or pseudoscientific beliefs used in the study were selected from the statements in “Prevalence of pseudoscientific beliefs” (Martí, Conover & Kidd, 2021), which collected representative prevalence estimates based on a large sample of over 900 Americans. Some statements included were widely known to be either true or false, such as ‘Plants need water to grow’ or ‘Dogs can fly’. Other statements were more ambiguous, like ‘The first baby born in the world in 1908 was a boy’ and ‘Will Smith likes grapes more than apples’. Some statements were uncertain, such as ‘People can maintain consciousness after death and visit living people’, and included a few conspiracy theories that may or not be true, such as ‘Powerful people planned the release of COVID-19 virus’.

Procedure/Design

Each participant was presented with 37 statements related to real-world conspiratorial or pseudo-scientific beliefs. For each statement, participants provided an initial likelihood rating (0-100) and a prevalence rating (0-100). They were then shown the same 37 statements in a shuffled order. This time, before providing new likelihood and prevalence estimates, participants were shown a sample indicating how many of 10 survey respondents endorsed each statement.

On half of the trials (statements), randomly assigned, this sample matched the participant’s own initial prevalence estimate. On the remaining trials, corresponding to the high prevalence condition, the number of people endorsing the pseudoscientific belief was set to be 40% higher than the participant’s initial prevalence estimate. Participants then provided a likelihood and prevalence estimate for each statement again.

18 participants were excluded for failing more than 1 of 4 attention checks, and an additional 19 participants were excluded for giving blank or unreasonable responses to more than 1 out of 6 bot-catch questions intermixed between experimental trials. This exclusion resulted in a final sample of 366 participants. In addition to these preregistered exclusions, trials in which participants who initially endorsed the empirically unsupported belief (Martí, Conover & Kidd, 2021) with a likelihood of over 60% were also excluded because the 40% exaggeration in the high prevalence condition could not be applied to these. These trials only constituted 13.1% of the data, and their removal did not significantly affect the interpretation of the paper’s (Orticio, 2021) analyses.

The dataset available for the present analysis includes 205 participants. For this analysis, only the high prevalence condition was considered. After filtering for this condition, the dataset

contains each participant's likelihood and prevalence estimates for 25 statements, measured both before and after viewing the high prevalence sample for each statement.

Data Analytic Plan

The influence of education on belief likelihood and prevalence change under high social prevalence and uncertainty

The first hypothesis examines whether the changes in a belief's likelihood and prevalence are predicted by participant's initial certainty and moderated by their education level. Here, the participant's **initial certainty for each statement** and their **education level** are **independent variables**, whereas the belief **likelihood and prevalence change** are the 2 **dependent variables**.

As a measure of certainty, participants' initial likelihood estimates for each belief were used to calculate their distance from 50% on the likelihood scale. Thus, higher values indicate more certainty whereas values closer to 0 indicate uncertainty.

The dataset used for the current analyses includes 8 distinct education levels: Bachelor, Some College, Professional, Associate, Primary School, High School, Masters and PhD. Because the numbers of participants across these education levels was imbalanced, the 8 categories were collapsed into 3 broader groups: School (Primary School and High School), Some College (Bachelor, Some College, Professional, and Associate), Masters and above (Masters and PhD). These 3 education levels were ordered as School, Some College, followed by Masters and above.

The influence of political views on belief likelihood and prevalence change under high social prevalence and uncertainty

Similarly, the second hypothesis examines whether the changes in a statement/belief likelihood and prevalence are predicted by participant's initial certainty and moderated by their political views. Here, the participant's **initial certainty for each statement** and their **political views** are **independent variables** whereas the belief **likelihood and prevalence change** are the 2 **dependent variables**.

There were 6 distinct political views represented in the dataset: Liberal, Very Liberal, Moderate, Other, Conservative, Very Conservative. Since the numbers of participants within each category was skewed, Liberal and Very Liberal were combined into a single Liberal category, and Conservative and Very Conservative were combined into a single Conservative category. This resulted in 4 political view categories: Liberal, Moderate, Other, and Conservative.

For both the first and second hypotheses, a cross-classified multilevel model was used to assess the relationship between initial certainty and belief change. The model included initial

certainty, belief likelihood and prevalence change, and education and political views as fixed effects, with participants and statements level variation as random effects.

All analyses were conducted using R software, with the psych, lavaan, lmerTest, sjstats, ggplot2, and interactions packages.

Results

Intraclass correlation coefficient

The intraclass correlation coefficient indicates that 9.5% of the variance in belief likelihood change is due to differences between participants, while 4.9% is attributed to differences between Statements. For prevalence change, 5.6% of the variance is attributable to differences between participants and 6.3% to differences between statements.

For likelihood change, a large proportion of variation is due to differences between participants than statements, whereas for prevalence change, the variance due to differences between statements is slightly higher compared to participants.

How does Education affect belief likelihood estimates depending on initial belief certainty?

It was hypothesized that participants with less education change their belief likelihood more under uncertainty, especially when the prevalence is high. Figure 1 shows the mean change in participant's estimates of belief likelihood after viewing new prevalence data for each belief, by Education level.

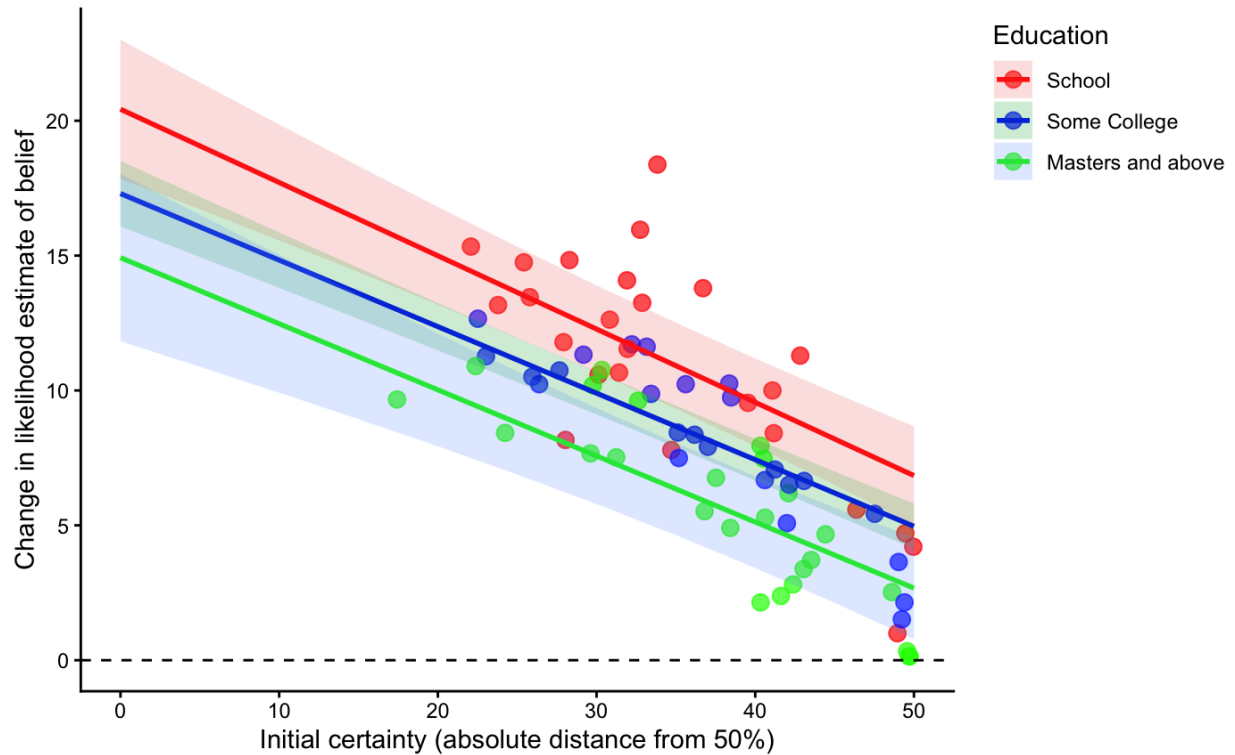


Figure 1: Mean change in participant's estimates of belief likelihood after viewing new prevalence data for each belief, by Education.

There is a significant difference in how much people change their belief likelihood, both at low and high initial certainty, based on education level. Participants with school level education showed the largest change (intercept = 20.43, $p < 0.001$), followed by those with some college level education (intercept=17.3, $p < 0.001$), and participants with masters and above (intercept=14.92, $p < 0.001$). This is in line with the hypothesis that participants with less education change their belief likelihood more when uncertain.

However, the effect of initial certainty on likelihood change ($\beta = 0.02$) did not meaningfully differ across education levels.

How does Education affect belief prevalence estimates depending on initial belief certainty?

Participants with less education were expected to adjust their belief prevalence more under uncertainty, particularly in high prevalence conditions. Figure 2 displays the mean change in belief prevalence for each belief, by Education.

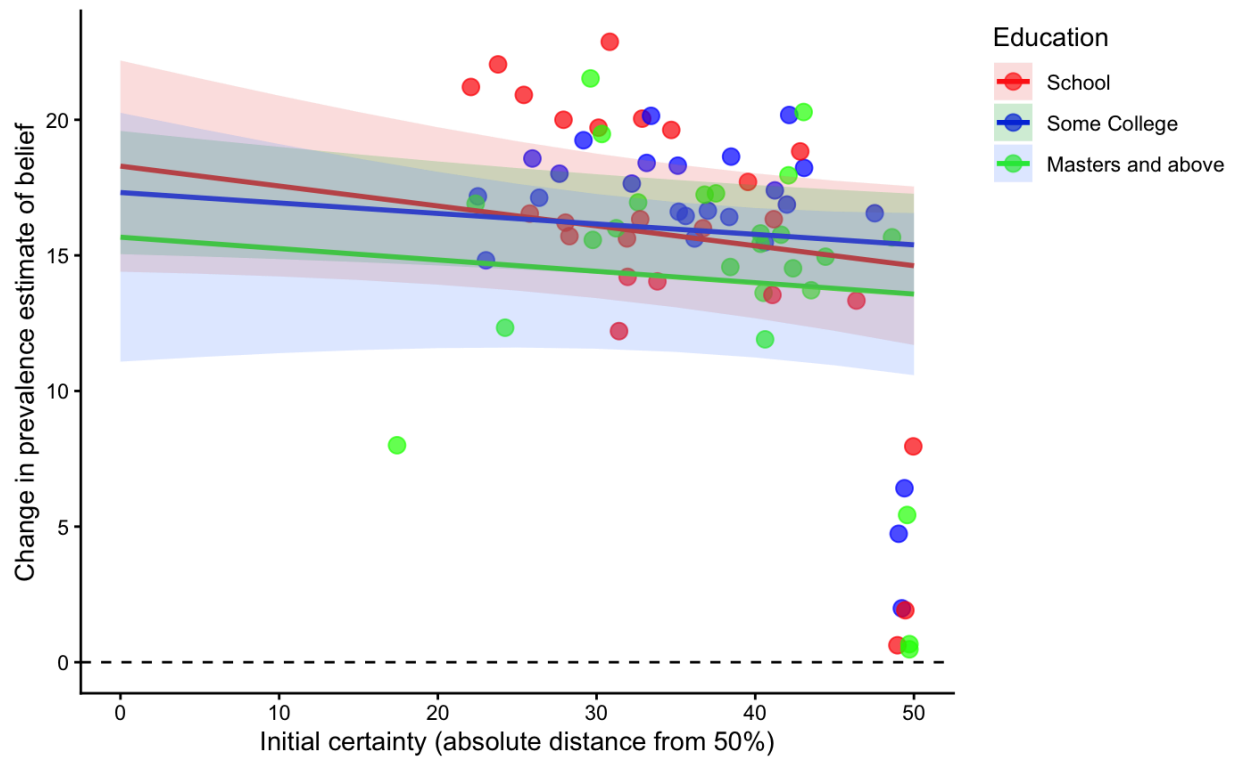


Figure 2: Mean change in participant's estimates of belief prevalence after viewing new prevalence data for each belief, by Education.

The results indicate that education strongly predicts baseline prevalence change. When participants were uncertain about a statement, those with master's degree or above showed the smallest change in prevalence (intercept=15.67, $p < 0.001$), followed by participants with some college level education (intercept=17.31, $p < 0.001$) and those with school level education (intercept=18.29, $p < 0.001$). These results are consistent with the hypothesis.

However, education did not significantly moderate the effect of initial certainty on prevalence change, since the effect ($\beta = 0.03$) is similar across all education levels.

How does Political view affect belief likelihood estimates depending on initial belief certainty?

It was hypothesized that participants with liberal political views would be less likely to adjust their belief likelihood less under uncertainty and high prevalence. Figure 3 displays the mean change in belief likelihood for each belief by political view.

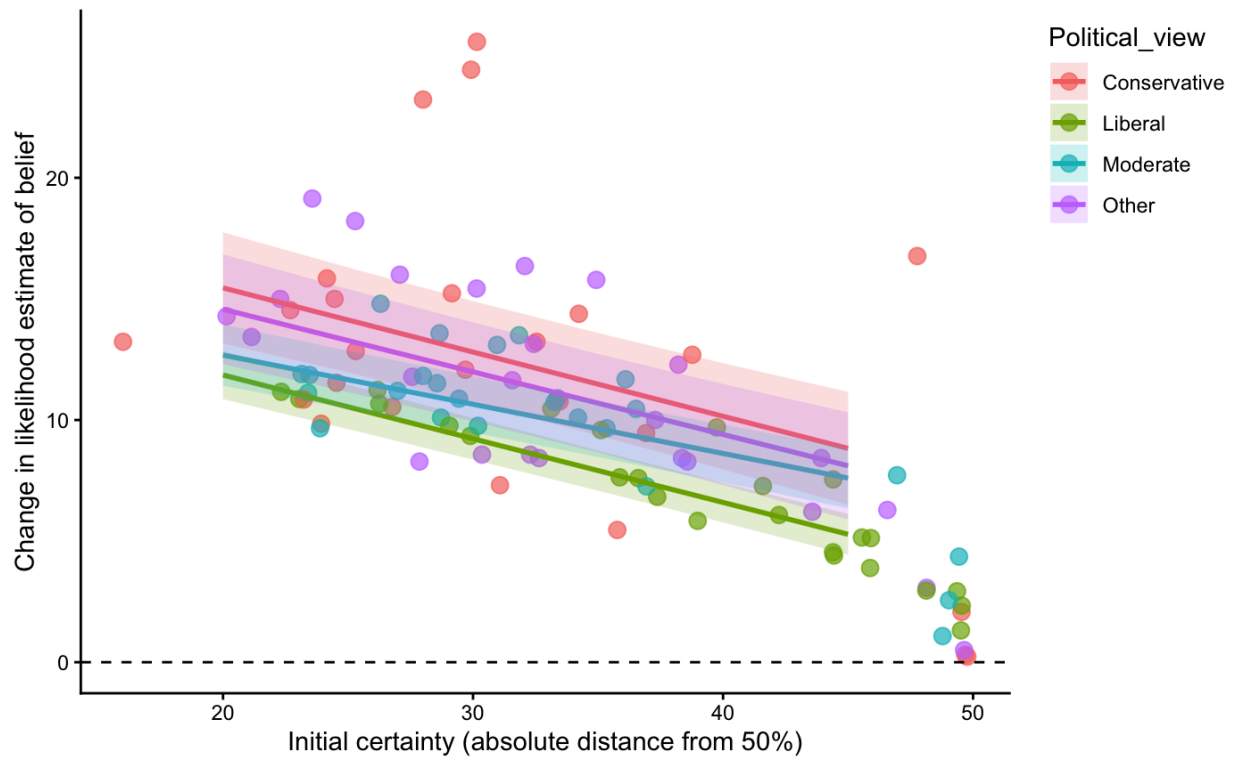


Figure 3: Mean change in participant's estimates of belief likelihood after viewing new prevalence data for each belief, by political view.

Across all political groups, participants with lower initial certainty exhibited greater belief updating. Baseline belief change differed by political orientation, with Conservatives (intercept=20.76, $p<0.001$) and participants categorized as "Other" (intercept=19.75, $p<0.001$) showing slightly larger changes than Liberals (intercept=17.12, $p<0.001$) and participants in the "Moderate" category (intercept=16.74, $p<0.001$). These results are consistent with the hypothesis that participants with liberal political views would change their belief likelihood less under uncertainty and high prevalence.

However, political view did not significantly moderate the relationship between initial certainty and belief likelihood change.

How does Political view affect belief prevalence estimates depending on initial belief certainty?

Consistent with the pattern observed for belief likelihood, it was hypothesized that participants with liberal political views would change their belief prevalence less under uncertainty and high prevalence. Figure 4 displays the mean change in belief prevalence for each belief by political view.

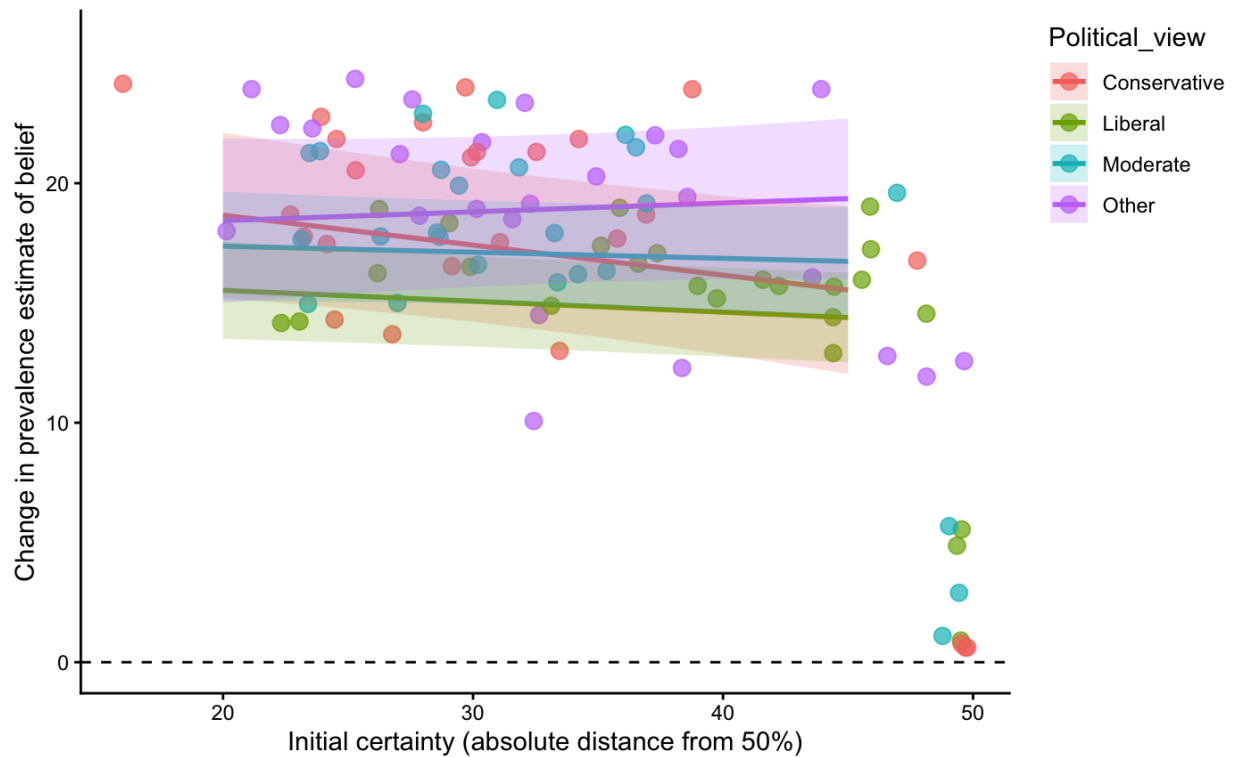


Figure 4: Mean change in participant's estimates of belief prevalence after viewing new prevalence data for each belief, by political view.

Across political orientations, lower initial certainty was associated with more changes in prevalence estimates ($\beta = -0.125$, $p < 0.05$). Baseline prevalence changes varied by political view, with Conservatives showing the largest change (intercept=21.18, $p < 0.001$) and Liberals showing the smallest change (intercept=16.43, $p < 0.001$) relative to other groups. Notably, there is a significant interaction between initial certainty and "Other" ($\beta = 0.16$, $p < 0.05$) political category, such that prevalence change increases with initial certainty, counter to the overall negative relationship.

Discussion

As predicted, the results demonstrate that education level influences the baseline belief likelihood and prevalence changes. Participants with school-level education showed greater belief updating compared to those with master's degrees or higher. Similarly, participants with liberal political views showed less belief change under uncertainty and high prevalence. However, no significant interaction was found between initial certainty and Education or initial certainty and Political view in predicting belief changes.

These results, derived from a subset of the original dataset, show promising trends that warrant further analysis using the complete dataset. Additionally, it would be interesting to explore statement-wise differences across various demographic factors.

References

Orticio, E., Marti, L., & Kidd, C. (2021). Beliefs are most swayed by social prevalence under uncertainty. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 43, No. 43).

Mart., L. Conover, A., & Kidd, C. (2021). Prevalence of pseudoscientific beliefs. Unpublished manuscript.

Additional results

Education and likelihood change

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method
[lmerModLmerTest]

Formula: likelihood_change ~ initial_certainty * Education - 1 + (1 | Statements) + (1 | ID)
Data: beliefs_data

AIC	BIC	logLik	-2*log(L)	df.resid
39700.7	39759.5	-19841.3	39682.7	5091

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.3527	-0.4550	-0.2178	0.1442	8.1862

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	9.002	3.000
Statements	(Intercept)	1.569	1.253
Residual		133.987	11.575

Number of obs: 5100, groups: ID, 204; Statements, 25

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
initial_certainty	-0.27166	0.02969	5020.10836	-9.150	<2e-16 ***
EducationSchool	20.42690	1.31277	1062.60416	15.560	<2e-16 ***
EducationSome College	17.29930	0.61805	264.42111	27.990	<2e-16 ***
EducationMasters and above	14.92382	1.57897	1350.60800	9.452	<2e-16 ***
initial_certainty:EducationSome College	0.02501	0.03175	5066.19378	0.788	0.431
initial_certainty:EducationMasters and above	0.02665	0.04530	4963.13278	0.588	0.556

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
      intl_c EdctnS EdctSC EdcMaa i_:ESC
EducatnSchl -0.785
EdctnSmCllg -0.045  0.113
EdctnMstraa -0.019  0.045  0.105
intl_cr:ESC -0.911  0.715 -0.277 -0.004
intl_c:EMaa -0.638  0.501 -0.008 -0.632  0.601
```

Education and prevalence change

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method

['lmerModLmerTest']

Formula: prevalence_change ~ initial_certainty * Education - 1 + (1 | Statements) + (1 | ID)

Data: beliefs_data

```
      AIC      BIC logLik -2*log(L) df.resid
43254.7 43313.6 -21618.4 43236.7    5091
```

Scaled residuals:

```
      Min      1Q  Median      3Q      Max
-1.6661 -0.8016 -0.2926  0.8785  5.8906
```

Random effects:

```
Groups   Name      Variance Std.Dev.
ID       (Intercept) 16.02    4.003
Statements (Intercept) 17.23    4.151
Residual                267.87  16.367
```

Number of obs: 5100, groups: ID, 204; Statements, 25

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
initial_certainty	-0.07345	0.04211	5064.46293	-1.744	0.0812 .
EducationSchool	18.29147	1.98493	492.68797	9.215	< 2e-16 ***
EducationSome College	17.31779	1.15811	77.39008	14.954	< 2e-16 ***
EducationMasters and above	15.67092	2.33920	776.11197	6.699	4.02e-11 ***
initial_certainty:EducationSome College	0.03488	0.04482	5046.58294	0.778	0.4365
initial_certainty:EducationMasters and above	0.03150	0.06393	4908.64157	0.493	0.6222

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
      intl_c EdctnS EdctSC EdcMaa i_:ESC
EducatnSchl -0.736
EdctnSmCllg -0.047  0.335
EdctnMstraa -0.026  0.167  0.295
intl_cr:ESC -0.906  0.667 -0.210 -0.005
intl_c:EMaa -0.634  0.467 -0.008 -0.603  0.601
```

Political view and likelihood change

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method

['lmerModLmerTest']

Formula: likelihood_change ~ initial_certainty * Political_view + (1 | Statements) + (1 | ID)

Data: beliefs_data

```
      AIC      BIC  logLik -2*log(L)  df.resid
39881.4 39953.3 -19929.7 39859.4    5114
```

Scaled residuals:

```
      Min      1Q  Median      3Q      Max
-2.2212 -0.4635 -0.2129  0.1507  8.2435
```

Random effects:

```
Groups   Name      Variance Std.Dev.
ID      (Intercept)  8.708    2.951
Statements (Intercept) 1.616    1.271
Residual                133.610  11.559
```

Number of obs: 5125, groups: ID, 205; Statements, 25

Fixed effects:

```
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)      2.076e+01 1.653e+00 9.730e+02 12.554 < 2e-16 ***
initial_certainty -2.650e-01 3.950e-02 5.075e+03 -6.710 2.17e-11 ***
Political_viewLiberal -3.637e+00 1.753e+00 1.060e+03 -2.075 0.0382 *
Political_viewModerate -4.013e+00 1.834e+00 9.957e+02 -2.188 0.0289 *
Political_viewOther -1.011e+00 2.303e+00 1.042e+03 -0.439 0.6608
initial_certainty:Political_viewLiberal 1.974e-03 4.183e-02 5.087e+03 0.047 0.9624
initial_certainty:Political_viewModerate 6.192e-02 4.406e-02 5.078e+03 1.405 0.1600
initial_certainty:Political_viewOther 6.464e-03 5.494e-02 5.080e+03 0.118 0.9063
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr) intl_c Pltc_L Pltc_M Pltc_O i_:P_L i_:P_M
intl_crtnty -0.762
PltcL_vwLbr -0.911 0.707
PltcL_vwMdr -0.873 0.678 0.823
PltcL_vwOth -0.695 0.540 0.657 0.627
intl_cr:P_L 0.709 -0.931 -0.781 -0.640 -0.511
intl_cr:P_M 0.674 -0.885 -0.636 -0.771 -0.484 0.835
intl_cr:P_O 0.541 -0.710 -0.511 -0.487 -0.779 0.671 0.636

Political view and prevalence change

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method

[lmerModLmerTest]

Formula: prevalence_change ~ initial_certainty * Political_view - 1 + (1 | Statements) + (1 | ID)

Data: beliefs_data

AIC	BIC	logLik	-2*log(L)	df.resid
43454	43526	-21716	43432	5114

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.6624	-0.8021	-0.2957	0.8769	5.9252

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	14.57	3.818
Statements	(Intercept)	17.42	4.173
Residual		267.59	16.358

Number of obs: 5125, groups: ID, 205; Statements, 25

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
initial_certainty	-0.12530	0.05588	5039.63774	-2.242	0.0250 *
Political_viewConservative	21.17565	2.41234	703.93974	8.778	< 2e-16 ***
Political_viewLiberal	16.43897	1.27684	110.12074	12.875	< 2e-16 ***
Political_viewModerate	17.88231	1.45892	181.75091	12.257	< 2e-16 ***
Political_viewOther	17.69736	2.41886	752.77052	7.316	6.54e-13 ***
initial_certainty:Political_viewLiberal	0.07976	0.05911	5040.67923	1.349	0.1773
initial_certainty:Political_viewModerate	0.09982	0.06218	5043.25700	1.605	0.1085

initial_certainty:Political_viewOther 0.16222 0.07754 5043.62676 2.092 0.0365 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	intl_c	Pltc_C	Pltc_L	Pltc_M	Pltc_O	i_:P_L	i_:P_M
PltcL_vwCns	-0.739						
PltcL_vwLbr	-0.033	0.251					
PltcL_vwMdr	-0.022	0.214	0.415				
PltcL_vwOth	-0.013	0.129	0.254	0.213			
intl_cr:P_L	-0.927	0.685	-0.221	-0.002	-0.003		
intl_cr:P_M	-0.883	0.652	0.001	-0.290	0.000	0.834	
intl_cr:P_O	-0.708	0.523	-0.002	0.001	-0.525	0.671	0.636